

DSA 104 Mock Exam: ML and AI in Chemistry

Name:

Matr. No.:

Signature:

Please read the questions carefully. Draw and write down **ONLY** what is necessary to answer the question in the space allocated underneath each question. You use some extra paper for your preliminary drawings or sketches (can be requested), only the stapled document will be corrected. Please do not use pencil, Tipp-Ex and/or red pens.

Task	Score
Definitions (Q1-2)	/ 12
Data (Q3-7)	/ 17
Basic ML (Q8-14)	/ 37
NNs (Q15-20)	/ 21
Critical thinking (Q21-23)	/ 14
Case study (Q24)	/ 14
Total	/ 115

- 1) What is *machine learning*? Give a **definition** and **one example application**. Name at least **three advantages of ML**. (5P)

ML: learning patterns from data, “rules” inferred from data; part of AI.

Application: Classification of plants according to their features, regression to predict the outcome of a process

Pros of ML:

- *Discovers patterns in data that would be missed by humans (many features!)*
- *(Relatively) fast, scalable*
- *Proven theory, well understood*
- *Results objective, quite repeatable*

- 2) **Give a definition** for each of the following terms. (7P)

Feature matrix:

Features = representation of an object (e.g. physical measurements, ...); Feature matrix = sum of all feature vectors of all samples. Dataset minus the things we want to predict (targets/labels)

Label:

Target data, classical term for target in classifications; data that we want to predict.

Classification:

Predicting categorical class labels, not numerical values

Empiric Risk Minimisation:

ERM estimates the generalization error (based on loss functions calculated from data) instead of the true probability of an error. Model parameters optimized to minimise generalisation error

Deep Learning:

Subset of ML, learns features from data (NNs!)

Halluzination: AI models produce seemingly correct, but actually untrue output.

3) Data: What are typically some key limitations with data in the life sciences? **Name two examples** and **state how each affects any ML models**. (4P)

- *Noisy: High degree of noise may limit prediction and cause wrong model choices (bias vs. variance!)*
- *Sparse data: Complex problems require large datasets, often not the case. Applicability of models needs to be estimated, generalisation poor.*

4) What is feature engineering? Give a brief **definition** and name at least **three fundamental tools!** (4P)

Process of enriching the dataset with features usable for machine learning: comprises creation, transformation, extraction, selection, scaling

Tools: e.g. scaling, filtering, encoding

5) What is scaling needed for? Why is it best implemented in a pipeline? (2P)

Data preprocessing step to treat all features with the same importance, regardless of their unit or order of magnitude (subtract mean, divide by std. dev.)

Used in pipeline to avoid data leakage: scaling factors only derived from test set!

6) **Explain** data leakage (in the technical sense, not in the media context), give **two examples**, and **name one fundamental practice** how to avoid data leakage. (4P)

Information from the test set leaks into the training process of a model – a violation of the principle that the test set (to determine generalisation on unseen data) is properly independent!

E.g. Using the test set for model evaluation and subsequent hyperparameter tuning, scaling on the entire dataset, duplicates of data in training and test set.

Good practice: Split data into training, validation and test data

7) **What is data bias?** State **two examples** where could they be introduced? (3P)

Data is not representing the ground truth accurately; distorted data that leads to skewed or unfair model behaviour.

Could be introduced at

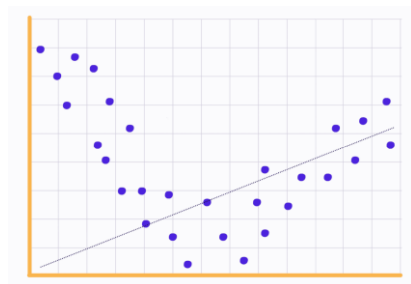
- *Sampling / Class imbalance (e.g. only a certain age group is surveyed)*
- *Feature extraction (e.g. cherry picking)*

8) **Explain** underfitting, provide an **exemplary graph** and discuss **how to resolve it!** (3P)

Underfitting: model is not sufficiently complex to explain the pattern in the data (=high bias), e.g.

How to deal with it?

Increase model complexity (e.g. depth of an NN, depth of trees, etc)



9) Consider the following datatypes and **assess how they are usually represented**, and **name a challenge that might be associated with each** of them. **Name an example application (regression/classification) for each** and describe **which ML model** would be used for them. (12P)

- Protein sequence

Representation: Text/String of e.g. single letter codes

Challenge: nonnumerical, order matters!

Task: Protein classification

Model: Transformers, RNNs

- Molecular structure

Representation: Strings, graphs

Challenge: 3D reality often lost in representations; non-numerical

Task: property prediction of small molecules (drug design!)

Models: Classic ML models via encoding, GNNs

- Process parameters (Temperature, pressure, etc.)

Representation: Numerical

Challenge: different units/scales

Task: forecasting, predicting reaction yields purely based on process parameters

Models: classic regressors, e.g. Linear regression, RF regressor, Ridge regressor

10) Why is accuracy alone not the best metric? **Explain!** Name at least **two more metrics** that could be used to judge a classification model. (3P)

Can give a misleading picture of model performance, e.g. when data is imbalanced or when different types of errors have different consequences (e.g. 95% of healthy species in training set, model will mainly predict healthy)

Only gives the number of true predictions divided by total predictions, no probability, etc.

Other metrics: e.g. ROC-AUC, precision, Recall/Sensitivity, F1 score,

11) **Explain** how a decision tree works. What is a **very common problem** of this model and **how can it be addressed**? What **advantages and disadvantages** does a **Random Forest** provide in comparison? **Name two each!** (7P)

The model starts at a root node with all data, selects a feature and threshold that best separates the data, splits into branches – iterate until reaching leaf nodes with the final prediction

Common problem: Overfitting, can be addressed e.g. by pruning (limiting depth of tree, ...) or using an RF instead (levels or randomization)

RF = ensemble of trees.

Pros: reduced overfitting (averages decisions of multiple trees with different decisions), higher predictive performance

Cons: lower interpretability, higher computational costs

12) **Explain Crossvalidation** and **why it is needed**. Name **two common methods** and **explain them briefly**. (6P)

CV: split data into multiple subsets (splits), train and test model on different splits, average results for more reliable performance estimate.

Why? Single train-test split can give misleading results; CV helps to reduce overfitting (averaging multiple splits)

Two methods:

- *k-fold CV: data split into k parts, trained on k-1, 1 split as test set; repeat k times (permute the test set), final result = average across all splits*
- *Leave-one-out CV: k=number of samples, train on all samples but one, permute through all samples (special case of k-fold CV)*

13) How to train your ~~dragon~~ model: Give some pseudocode for a training pipeline of any ML model. (3P)

- a) *Data preparation (cleaning, scaling)*
- b) *Train-(validate)-test split*
- c) *Model selection / initialise model*
- d) *Training: fit the model*
- e) *Validation (CV for hyperparameter tuning)*
- f) *Evaluation on test set for generalisation assessment (detect overfitting, ...)*

14) **Explain Exploration and Exploitation. Give an example** of where these principles may find application. (3P)

Exploration: try new things under the assumption that they may be good

Exploitation: Go for the highest reward, go for actions that the model already knows will be favourable

Classic case: Reinforcement learning

15) **Explain** how an artificial neuron **calculates its output from its inputs**. Explain **activation**. (3P)

- *Takes feature vector and calculates a weighted sum plus bias ($z = wx + b$)*
- *Applies activation function based on z ; if activation potential is passed, produce output*

Activation: if a certain potential is passed, a certain function will be applied, e.g. ReLU returns 0 for $z < 0$, but $y = z$ for $z > 0$; introduce non-linearity!

16) What is an MLP / FNN? Give a **brief definition** and explain at least **three of its key components**. (4P)

Multilayer perceptron / forwardfed NN = basic NN type, using a couple of hidden layers (depth!) between input layer (accepts raw data) and output layer (produces predictions), each hidden layer with a certain number of neurons (width), learning non-linear trends.

Components:

- *layers (see above)*
- *forward pass: how the data passes through the NN from the input data to the prediction outcome*
- *Backpropagation: after forward pass, loss is calculated, then compute gradients for weight optimization working backwards from output to input.*

- 17) **Explain the components** of a convolutional layer. Give **an example** where a CNN might be applied in the life sciences. (3P)

Convolutional layer = kernel application ("filter the input data") + activation function.

Kernel = weight matrix / receptive field passed over the data, performing a certain operation

Activation function (see above)

Application: e.g. cell counting in microscopy

- 18) State **one reason** why transformer models are dominant for sequence learning? Name at least **one other model class** that may be applicable there and why these models are at a **disadvantage** compared to transformers. (3P)

Transformers excel in sequence data (text, ordered sequences of characters) by applying self-attention to calculate weights between all tokens to determine their relevance. Processes entire sequence at once, not sequentially

Contrast: RNNs process sequences token by token, builds up memory (different models, e.g. LLSTMs, GRUs)

Disadvantages of RNNs: Sequential learning slow, memory limited (tend to forget without any gates, ...)

- 19) What is transfer learning? Give a **definition! Name and explain** in short **two approaches**. (5P)

A model is trained on a source task (e.g. general images), the knowledge is then used to improve another, related task (target task, e.g. cell microscopy images).

Linear probe / Feature extraction: a pretrained model is used only for feature extraction / encoding (weights of this remain unchanged), another model uses these features as input to the prediction task

Fine-tuning: Model is pretrained on a source data set (large and wide!), then this pretrained model is refitted to the target task by unfreezing layers (updating some weights). The fine-tuned model is then used for prediction.

20) Explain in short how a GNN works (3P)

A GNN operates on graph data (nodes and connecting edges):

- *Each node starts with feature vector (e.g. atom properties)*
- *Model updates each node by aggregating information from neighbors*
- *Repeated over several layers (message passing) so nodes capture local and global structure*
- *Final node or graph representations are then used for prediction (e.g. classification)*

21) Name **three key ethical aspects** of machine learning (3P)

Transparency: Black box approach limits traceability of data and how outcomes are achieved. Lack of ensuring fair usage, respecting IP, ...

Intellectual property: Who owns the rights to generated data? Were copyrights infringed in this creation by leaked IP into training data? Sensible data used as training data. How do the developing tech companies for big LLMs handle user data?

Halluzinations: Can we actually believe what a model is predicting? Some outcomes may seem perfectly fine, but would never survive any fact check...

Misuse: Deepfakes, using ML models to “guess” passwords, generated research outcomes marketed as real ones...

Fairness: Are models systematically disadvantaging certain population groups (e.g. ethnicity, gender, ...)? Are “open” models actually available to everyone? ...

22) **Explain** applicability domains and **name and explain at least two different AD metrics** and explain in short. (5P)

Applicability Domain = the region of descriptor space where a machine learning model is reliable and makes trustworthy predictions (v.v. for outside of the AD). Models are good at interpolation, worse at extrapolation!

Metrics:

- *Distance based (e.g. kNN): Measure how far a new sample is from training data points; if too far away, it is considered outside the applicability domain*
- *Density based (KDE): Estimate how densely populated the region around a sample is; Low-density regions = fewer training examples = unreliable predictions*

23) Critical thinking – comment on the following statements: (6P)

- “More complex models and a high amount of data lead to better insights/predictions”

Quantity is not quality! High amounts of high-quality data is definitely a good thing.

Higher model complexity does not necessarily mean better results, completely depends on task (compare RF vs. Pretrained NNs in simple prediction tasks). More complex models tend to overfit!

- “ChatGPT suggested a new catalyst (SMILES) and it actually worked. It is great to see that the finally implemented some chemistry knowledge in LLMs!”

No knowledge implemented, ChatGPT has nothing of that sort, no agents, etc.. Just based on the available training data, the predictions have improved. Since ChatGPT is an LLM it is good at text-based tasks, generating new text (SMILES strings) is easy when you have data to mimic. This has nothing to do with knowledge – although some rules may be inferred from the data (DL!). The outcome resembles a valid string – but could just as well be hallucinations.

- “With the progress in AI and automation, soon, there will no longer be a need for human scientists.”

Displacement instead of replacement. Role of the human should be even more pronounced the more automation and AI are applied, even more crucial! Alas, what good is the best scientific output if no scientist is there to use it / read it / understand it?

AI tools are still mainly statistical machines, automation may give them a physical body, but they lack some of the most important driving forces behind innovation and science: compassion, creativity, ethic considerations, ...

24) Case study: You are supposed to predict the activity of some small molecules against a certain cancer cell line.

a) Propose a high-level ML pipeline (data → model → validation → outcome). **Explain each part** in short and describe what **considerations for each step** may be valid (10P)

- *data preparation, cleaning, scaling: Clean data needed for ML, target needs to be separated from features, features need. Considerations: Scaling needed (e.g. not for RF)? Removal of NaNs might be critical for the model? How to treat duplicates and outliers, ...*
- *data split: train-test split with random seed for reproducibility. Crucial to avoid data leakage. Classic 80 % training data. Considerations: Stratify (classifications)? Random splits? Validation set needed or CV done on training data?*
- *Model choice and training: depending on the task! Regression models to be chose here – e.g. compare different regressor classes (or go for a solid baseline with a RF regressor). Model hyperparameters are to be tuned with CV (e.g. k-fold CV), depending on the size of the data set, smaller models may offer a time-advantage in training and CV (limit parameters, etc.). Use best model to train on the training set, choosing an appropriate metric (e.g. MSE to punish large deviations)*
- *Evaluate generalisation by running the model on test set. Overfitting to be assessed by comparing training and test performance. Considerations: Which metric best to use?)*
- *Deploy the model on unseen data and predict. Ideally evaluate experimentally!*

b) Identify two major challenges specific to this task. (2P)

Dataset may be small, chemical space coverage of known molecules may be insufficient, data may be very noisy (human assays...), data may come from very different assays or labs!

c) Explain how you would evaluate whether the model is clinically useful. (2P)

Reference systems (similarity to unseen models). Determine applicability domain, quantify uncertainty (e.g. ensemble models).

Promising candidates should always be tested experimentally = only way to assess the ground truth!